# TACTILE-LANGUAGE GROUNDING DEBT: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

**Anonymous authors**
Paper under double-blind review

## ABSTRACT

We study tactile-language-action models. The motivating bet is: Measure what language-conditioned policies fail to ground until touch resolves it.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

## 1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For tactile-language-action models, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

## 2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2024a;b;c; 2023; 2026; 2024d). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

## 3 MECHANISM

Given state $s$ and action $a$, the model returns an observed next-state prediction plus a finite atlas $B(s,a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

## 4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

| Variant | Split | Success | Tail risk |
|---|---|---|---|
| observed_only_wam | nominal | 0.557 | 0.443 |
| observed_only_wam | hidden_mode_shift | 0.180 | 0.820 |
| augmentation_as_data_wam | nominal | 0.598 | 0.402 |
| augmentation_as_data_wam | hidden_mode_shift | 0.273 | 0.728 |
| scalar_uncertainty_wam | nominal | 0.600 | 0.400 |
| scalar_uncertainty_wam | hidden_mode_shift | 0.312 | 0.688 |
| proposed_branch_mechanism | nominal | 0.812 | 0.188 |
| proposed_branch_mechanism | hidden_mode_shift | 0.575 | 0.425 |

## 5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

## 6 CONCLUSION

Tactile-Language Grounding Debt argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

## REFERENCES


Dragon: A dialogue-based robot for assistive navigation with visual language grounding, 2023. http://arxiv.org/abs/2307.06924v3.

Lingo-space: Language-conditioned incremental grounding for space, 2024a. https://doi.org/10.1609/aaai.v38i9.28898.

Negative object presence evaluation (nope) to measure object hallucination in vision-language models, 2024b. https://doi.org/10.18653/v1/2024.alvr-1.4.

Saynav: Grounding large language models for dynamic planning to navigation in new environments, 2024c. https://doi.org/10.1609/icaps.v34i1.31506.

What matters in building vision-language-action models for generalist robots, 2024d. http://arxiv.org/abs/2412.14058v4.

When vision overrides language: Evaluating and mitigating counterfactual failures in vlas, 2026. http://arxiv.org/abs/2602.17659v1.